JOSÉ L. RAMÍREZ-COLÓN

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PROFESSIONAL SUMMARY

Interdisciplinary scientist with a Ph.D. and 9+ years of research experience spanning molecular and computational science, analytical chemistry, planetary science, and scientific instrumentation. Experienced in leading and coordinating multidisciplinary research, translating complex scientific concepts into clear deliverables, and communicating technical results to diverse audiences. Strong scientific communicator interested in roles at the intersection of science, technology, people, and cross-functional collaboration. Authorized to work and live in the EU.

EDUCATION

Georgia Institute of Technology (Atlanta, GA)
Ph.D., Earth and Atmospheric Sciences — Aug. 2021 - Jul. 2026
Minor, Microbial Genomics and Structural Biology
Graduate Certificate, Astrobiology

University of Puerto Rico, Río Piedras (San Juan, Puerto Rico)
B.S., Biology, Cellular and Molecular Biology track, *with honors* — Sep. 2017 - May 2021

RESEARCH & PROFESSIONAL EXPERIENCE

Research Scientist II — Aug. 2026 – Present
Planetary Exploration Laboratory, Georgia Institute of Technology

- Investigate how environmental conditions and molecular charge states influence amino acid electronic properties and their implications for biosignature detection and the genetic code.
- Use quantum-chemical modeling and statistical analysis to examine molecular reactivity across planetary conditions of astrobiological-relevance.
- Contribute to research deliverables, scientific figures, manuscripts, and proposal development across multidisciplinary laboratory projects.
- Mentor undergraduate researchers in research methods, data interpretation, and scientific communication; support laboratory coordination and group activities.

NSF Graduate Research Fellow/Graduate Research Assistant — Aug. 2021 – Jul. 2026
Planetary Exploration Laboratory, Georgia Institute of Technology

- Developed LUMOS, a computational and statistical framework integrating molecular electronic properties and chemical abundance data to distinguish patterns across biological and abiotic systems.
- Designed and applied computational chemistry workflows to evaluate molecular properties across diverse chemical structures and environmental conditions.
- Analyzed ground and parabolic flight single-molecule detection data from the Electronic Life-detection Instrument for Enceladus/Europa (ELIE).
- Assessed gravitational effects on nanogap sensing under microgravity and reduced-gravity conditions.
- Coordinated research across Georgia Tech, NASA Goddard, Osaka University, and the Smithsonian, contributing to scientific planning, data interpretation, and research deliverables.
- Contributed scientific writing, preliminary data, and figures to NASA and NSF proposal development; prepared publications and conference presentations.

Visiting Researcher Apr. 2024
Astrobiology Analytical Laboratory, NASA Goddard Space Flight Center

- Extracted and quantified amino acids from Murchison meteorite and Atacama Desert analog samples using cleanroom protocols and LC-MS.
- Generated enantiomeric reference measurements supporting collaborator experiments on chiral discrimination using single-molecule nanogap sensing.

Undergraduate Researcher Oct. 2017 – May 2021
WARM^2 Laboratory, University of Puerto Rico, Río Piedras

- Developed electrospun membranes and extraction protocols for targeted compound recovery from environmental water samples, with material characterization via SEM, FTIR, and UV-Vis.
- Developed solid-phase extraction (SPE) protocols to isolate and recover trace organic compounds from complex aqueous matrices.
- Contributed to a first-author ACS Omega publication and presentations at national and Puerto Rican scientific meetings.

Research Associate in Astrobiology May 2019 – Jul. 2019
Astrobiology Analytical Laboratory, NASA Goddard Space Flight Center

- Developed and validated a GC-MS method for trace amide detection in carbonaceous meteorites, optimizing extraction and derivatization to minimize matrix interference.

High School Intern Aug. 2016 – Dic. 2016
Ramos Laboratory, University of Puerto Rico, Humacao

- Fabricated activated carbon from *Crescentia cujete*'s shells to remove copper from water.
- Designed a 3D-printed filter and evaluated adsorption efficiency using atomic absorption spectroscopy.

LEADERSHIP & SCIENTIFIC COMMUNICATION

President Jun. 2025 – May 2026
ExplOrigins Student Organization, Georgia Institute of Technology

- Led year-long programming for a multidisciplinary astrobiology organization, organizing 12 scientific, professional-development, policy, and public-engagement events.
- Coordinated speakers, participants, university partners, and community organizations while managing event logistics and programming within a sub-$500 annual budget.
- Designed social media content and promotional materials; facilitated discussions with early-career scientists.

Science Communication Intern Aug. 2024 – Dec. 2024
Ciencia en tus Manos Inc.

- Conceptualized and launched *Lunes de Ciencia Gufiá*, a Spanish-language video initiative translating peer-reviewed research by Puerto Rican scientists into accessible scientific content.
- Produced six episodes, managing literature research, story selection, scientific review, scripts, visual development, recording, editing, and publication.

Social and Cultural Chair Jan. 2022 – Jan. 2023
Latino Organization of Graduate Students, Georgia Institute of Technology

- Planned cultural and social activities, including Hispanic and Latinx Heritage Month trivia and community events.
- Organized a discussion with Serve-Learn-Sustain on integrating United Nations Sustainable Development Goals into graduate research.

Team Leader, Science & Research Branch Aug. 2020 – Jun. 2021
RockSat-X Suborbital Rocket Mission, University of Puerto Rico, Río Piedras

- Led scientific activities for a student payload project and represented the science team in mission progress meetings.
- Developed and delivered technical flight-readiness presentations; contributed to mission success criteria, contamination mitigation, and scientific planning.
- Performed an end-to-end Lambda DNA control sequencing workflow using Oxford Nanopore MinION Mk1B; supported experimental preparation involving VolTRAX.

Treasurer Aug. 2018 – Jun. 2020
American Society for Biochemistry and Molecular Biology Student Chapter, University of Puerto Rico, Río Piedras

- Supported the planning and coordination of the virtual Seventh Annual ASBMB Research Symposium.
- Conducted outreach introducing elementary, middle, and high school students to careers in bioscience.
- Coordinated membership registration and maintained records for new members.

President Aug. 2018 – Jun. 2020
Astrobiology Circle, University of Puerto Rico, Río Piedras

- Participated and staffed a table at the First Puerto Rican Space Exploration Festival.
- Participated in the International Astronomical Union 100 NameExoWorlds global campaign.
- Conducted activities on micrometeorite collection and analysis.

TEACHING & COMMUNITY ENGAGEMENT

Graduate Teaching Assistant Jan. 2026 – May 2026
EAS 1601 Habitable Planet Lab, Georgia Institute of Technology

- Led two undergraduate laboratory sections serving more than 40 students, covering planetary systems, stellar evolution, climate, exoplanets, and habitability.
- Facilitated hands-on and project-based learning; guided teams through simulated exoplanet mission proposals.
- Provided grading and scientific communication feedback.
- Course Instructor Opinion Survey (CIOS) evaluations reported an interpolated median of 4.92/5.00 for subject familiarity and an overall instructor effectiveness score of 4.82/5.00.

Astrobiology Fellow May 2023 – May 2024
Astrobiology Program, Georgia Institute of Technology

- Coordinated interdisciplinary programming, including an external-speaker event and the ExplOrigins Colloquium.
- Managed event communications and designed promotional flyers.

Instructor & Mentor Apr. 2023 – Sep. 2023
iStar Class Research Program, IPERC Academy

- Mentored a South Korean high school student through a 20-week astrobiology research project on Mars habitability and agriculture.
- Guided literature review and scientific writing for a paper on Martian soil composition and crop suitability, published in the UCI Gifted and Talented Institute Science Journal in 2023.

Outreach Fellow Jun. 2018 – Jun. 2021
CIRE^{2}N Outreach Program, University of Puerto Rico, Río Piedras

- Worked with Caras con Causa and the Río Piedras Urban, Community, and Business Action Center on science outreach for elementary students and the Vietnam community in Cataño, Puerto Rico.

PUBLICATIONS

1. **Ramírez-Colón, J.L.**, Ziqin, N., Carr, C.E. (2026). *Distinguishing life from non-life via molecular frontier orbital energy gaps.* Preprint: arXiv:2602.18490. (*Under revision at Communications Chemistry*)
2. Ohshiro, T., Komoto, Y., Takaai, T., Yoshida, T., Washio, T., **Ramírez-Colón, J.L.**, McLain, H.L., Duzdevich, D., Zuber, M.T., Carr, C.E., Taniguchi, M. (2026). *Chiral discrimination of amino acids in meteorite and desert soil extracts via single-molecule nanogap conductance.* Nature Communications 17, 10274, doi.org/10.1038/s41467-026-77947-6.
3. **Ramírez-Colón, J.L.**, Johnson, E., Duzdevich, D., Lee, S., Soderblom, J., Zuber, M. T., Taniguchi, M., Ohshiro, T., Komoto, Y., and Carr, C. E. (2024). *Nanogap Solid-State Single-Molecule Detection at Mars, Europa, and Microgravity Conditions.* 2024 IEEE Aerospace Conference, Big Sky, MT, USA, 1-10, doi.org/10.1109/AERO58975.2024.10521175.
4. Carr, C.E., **Ramírez-Colón, J.L.**, Duzdevich, D., Lee, S., Taniguchi, M., Ohshiro, T., Komoto, Y., Soderblom, J. S., and Zuber, M. T. (2023). *Solid-State Single-Molecule Sensing with the Electronic Life-detection Instrument for Enceladus/Europa (ELIE).* Astrobiology, 23(10), 1056-1070, doi.org/10.1101/2022.08.31.505913
5. **Ramírez-Colón, J.L.**, Santiago-Maldonado, X., Laboy-López, S., Méndez Fernández, P.O., Torres-Díaz, M., Lasalde-Ramírez, J.A., Díaz-Vázquez, L.M., and Nicolau, E. (2022). *Porous Cellulose Acetate/Block Copolymer Membranes for the Recovery of Polyphenolic Compounds from Aquatic Environments.* ACS Omega, 7(3), 2774–2785, doi.org/10.1021/acsomega.1c04327.

Full publications list: Google Scholar

PRESENTATIONS

1. Electronic Robustness in Amino Acids: A Clue to Genetic Code Diversification. *Origins of Life Meeting 2026*, Georgia Institute of Technology (Sept. 22, 2026).
2. Environmental Robustness of Amino Acid Electronic Properties: Implications for Biosignature Detection Across Diverse Planetary Conditions. *2026 Astrobiology Science Conference (AbSciCon)*, Madison, Wisconsin (May 18, 2026). Abstract 115-008.
3. Mind the Gap: Distinguishing Biotic and Abiotic Chemistries via Frontier Orbitals in Amino Acids and Peptides. *2025 European Astrobiology Network Association Conference*, Lisbon, Portugal (Oct. 22, 2025). Abstract EANA2025-100.
4. Frontier Orbital Metrics Discern Abiotic & Biotic Amino Acid Systems. *2025 Astrobiology Graduate Conference (AbGradCon)*, CU Boulder (Jun. 11, 2025).
5. Towards the Search for Prebiotic Nucleosides in Carbonaceous Chondrites. *2024 Astrobiology Science Conference (AbSciCon)*, Providence, RI (May 9, 2024). Abstract 427-137.
6. Nanogap Solid-State Single-Molecule Detection at Mars, Europa, and Microgravity Conditions. *2024 IEEE Aerospace Conference*, Big Sky, MT (Mar. 6, 2024). Abstract 2836.
7. Single-molecule Detection of Amino Acids Using Solid-state Nanogap Sensing. *2022 Astrobiology Science Conference*, Atlanta, GA (May 19, 2022). Abstract 1030024.
8. Porous Electrospun Fiber Mats for the Recovery of Polyphenolic Compounds from the Cucharillas Marsh in Puerto Rico. *39th Puerto Rico Interdisciplinary Scientific Meeting 54th ACS Junior Technical Meeting*, Virtual (Apr. 23, 2021).
9. Silylation Protocol for the Analysis of Meteoritic Amides. *American Geophysical Union Fall Meeting*, Virtual Poster. Abstract 661596.

10. Porous Electrospun Fiber Mats for the Recovery of Polyphenolic Compounds in Water. *SACNAS Conference*, Virtual Poster. Abstract 65220.

11. Towards an Effective Method for the Analysis of Meteoritic Amides. *Europlanet Science Congress*, Virtual Poster. Abstract EPSC2020-150.

12. Development of electrospun microfibers for the recovery of polyphenolic compounds: A sustainability model. *Emerging Researchers National (ERN) Conference in STEM Annual Meeting*, Washington D.C. (Feb 7, 2020). Abstract 50.

13. Development of pH Responsive Cellulose Nanofibers for the Selective Adsorption of Emerging Organic Contaminants in Water. *2019 ACS Orlando Annual Meeting*, Orlando, FL (Mar. 31, 2019). Abstract 3110086.

HONORS AND AWARDS

2026 Network for Life Detection Travel Award to attend AbSciCon 2026.

2025 EANA Space Factor competition, Fourth Place.

2025 John Bradshaw Award for instrumentation proficiency.

2024 NASA SCoPE AGU SciAct Affiliate.

2023 Georgia Tech Astrobiology Fellowship.

2023 EAS Graduate Student Service Award, Georgia Tech.

2023 Herbert P. Haley Fellowship, Georgia Tech College of Sciences.

2023 National Science Foundation Graduate Research Fellowship Program Awardee.

2021 Georgia Tech President's Fellowship.

2021 AAAS Emerging Researchers National Conference in STEM Oral Session Award.

2018 CREST-CIRE^{2}N Fellowship, University of Puerto Rico.

2017 Second Prize for International Participants in Environmental Science and Engineering, CASTIC.

2017 Fourth Grand Award in Environmental Engineering, Intel ISEF.

2016 North America Community Impact Finalist, Google Science Fair.

SKILLS

Comp. chemistry	Rowan, GAMESS, Psi4, MOPAC, ChemDraw, Chem3D, Orca, PyMOL
Molecular modeling	Geometry opt., thermochemistry, conformer search, electronic properties
Analytical chem.	GC-MS, LC-MS, derivatization, chiral analysis
Molecular inst.	Nanogap sensing, Oxford Nanopore MinION, NanoDrop, Qubit
Materials char.	Electrospinning, SEM, EDS, FTIR, UV-Vis, tensile testing
Data analysis	MATLAB, OriginLab, Excel, Google Colab
Communication	Scientific writing, science communication, public speaking, mentorship
Design	Adobe Illustrator, Canva, BioRender
Languages	Spanish (native) & English (fluent)

WORKSHOPS AND SUMMER SCHOOLS

2025 NASA SCoPE Summer Symposium, Tempe, Arizona.

2025 Proposal Writing Retreat, AbGradCon, Nederland, Colorado.

2024 Culturally Inclusive Planetary Engagement, Bradley Observatory, Atlanta.

2024 Observatoire de Haute-Provence Summer School for Sample Returns, France.

2024 ComSciCon Atlanta.

OUTREACH ACTIVITIES

Mar 2026 *Guest speaker*, The Archipelago That Taught Me to Look Up, Global Journeys Course, Agnes Scott College, Decatur, Georgia.

Mar 2026 *Volunteer*, How Big is the Solar System, Really?, Atlanta Science Festival Celebrate STEAM, Atlanta, Georgia.

Nov 2024 *Volunteer*, Paper Strip Scale Model, Center for a New Generation at the Tuskegee Airmen Global Academy Boys & Girls Club, Atlanta, Georgia.

Nov 2023 *Invited Speaker*, Growing Plants on the Moon and Mars, The SAE School, Mableton, Georgia.

Apr 2023 *Contributor*, Abundance Distribution of Organic Carbon and Carbon Redox States, Life Detection Knowledge Base.

Mar 2023 *Volunteer*, How Big is the Solar System, Really?, Atlanta Science Festival, Atlanta, Georgia.

May 2022 *Spanish Interpreter*, Regeneron International Science and Engineering Fair, Atlanta, Georgia.

May 2021 *Contributor*, STEAM Education and Outreach during COVID19 Pandemia, Public Choice Award, 2021 STEAM FOR ALL VIDEO SHOWCASE: Covid, Equity, and Social Justice.

Apr 2021 *Coordinator*, Careers in STEAM, Today's Kids Montessori School, Virtual outreach.

Jan 2021 *Invited Speaker*, Tras el Camino de la Investigación, CIMATEC, Virtual outreach.

Dec 2020 *Panelist*, CIRE^{2}N Fellows Interview, CIRE^{2}N Nanoday 2020, Virtual interview.

Dec 2020 *Video Producer*, Graphene: The Material of the Future, CIRE^{2}N Fall Nanoday, Virtual outreach.

Nov 2020 *Coordinator*, Careers in STEAM, Homeschool Kids and Fun, Virtual outreach.

May 2020 *Activity designer*, Nanotechnology and COVID19, CIRE^{2}N Spring Nanoday, Virtual outreach.

Oct 2019 *Volunteer*, Molecular Sciences Research Center's Open House 2019, San Juan, Puerto Rico.

May 2019 *Volunteer and activity designer*, Utilizing self-assembled polymers for targeted drug delivery, Urban, Community and Business Action Center of Capetillo, Río Piedras, Puerto Rico.

Apr 2017 *Presenter*, Contaminación Hídrica, SCIENCE LIVE! del EcoExploratorio, Plaza Las Américas, Puerto Rico.